actually rises to the level of the upper border of the first ribs, while the innominate, subclavian, and common carotid on the right side and the subclavian and common carotid on the left side course the neck. During systole the pulsations of these large vessels cause the neck to expand elevating the chain and during diastole the collapse of these vessels is associated with the dropping of the chain. These vessels are so proximal to the heart as to make the pulsations practically synchronous with the cardiac cycle and the timing of cardiac murmurs is accurate.

SUMMARY

A thin necklace chain lightly suspended about a patient's neck serves as an ideal method for accurately timing cardiac murmurs.

MECHANICAL SUFFOCATION DURING INFANCY

A Comment on its Relation to the Total Problem of Sudden Death

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ABRAMSON,[1] in a recent contribution, has brought to attention accidental mechanical suffocation as a cause of death during infancy and has pointed out the alarming frequency with which this diagnosis is made. He has stated that one hundred and thirty-nine babies were reported dead from suffocation during a five-year period in New York City alone. This is a larger number than the total for measles, scarlet fever, and diphtheria and is as striking as the earlier report of Gafafer[2] who found suffocation to account for over one-third of all accidental deaths in infants under 1 year of age. It is difficult to read a city newspaper over an extended period of time and not be impressed by the frequency with which infants appear to be asphyxiated in their beds.

There is no doubt as to the seriousness of the problem but we wonder whether emphasis is being placed at the right point; a distressingly large number of babies die without noticeable predisposing cause, but is the primary premise correct, that all reported as having suffocated actually died from simple occlusion of air passages by bedding, sleeping attire, or other mechanical means? A certain number are garroted by tightly fitting sleeping devices or are compressed by larger bedfellows but we are far from convinced that good evidence exists to support the theory that more infants suffocate than die from burns, trauma, foreign bodies, and automobile accidents combined.[2]

One of the most distressing features to be faced upon the death of a child is the invariable presence of self-incrimination and guilt on the part of the parents. This can be dispelled only through sincere assurance by the attending physician that all possible was done to avert the catastrophe and that the outcome could not have been altered through changes in their behavior. To leave the family with a clear conscience is a duty secondary in importance only to saving the patient, and the physician should take all measures to be able to do this; it is often the prime motive for a post-mortem examination. It is there-

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fore in keeping that we should be overly critical of a diagnosis which saddles the family with the entire blame for the death of their baby; for having allowed him to smother.*

Just what evidence can be adduced for this belief that a large number of healthy infants die from suffocation? So far as we can determine from reading, talking to pathologists, and querying coroners, a large portion of it rests on folklore; upon such practices as keeping the cat off the bed for fear it might "steal the baby's breath." We find no record of anyone having observed an infant suffocate without benefit of patented sleeping devices or other mechanical aids, nor can we find work which might suggest that anoxemia has been experimentally produced through arrangements of the usual bedding in relation to a normal baby. Certainly there is nothing pathognomonic of suffocation to be seen at autopsy; a few petechial hemorrhages scattered over the pleura and reflecting at most an increased vascular permeability during extremis. Our everyday experiences, such as the difficulty of maintaining high oxygen concentration in fabric tents constructed for this purpose, certainly do not lead to a belief that flimsy bedclothes would be an effective deterrant to free gaseous diffusion.

Some time ago, in association with Dr. Robert McCammon, we analysed the atmosphere breathed by infants covered in various manners by different types of bedding. We were able to demonstrate a reduction in oxygen or increase in carbon dioxide only after the addition of a rubber sheet, secured tightly at each border. The infants under ordinary bedding showed no discomfort until sufficient time had elapsed for heat and humidity to come into play. We also attempted to induce anoxemia by having the subjects sleep with nose and mouth closely approximated to mattresses and pillows but here again we were unsuccessful since the smallest was capable of rolling to obtain an airway and the larger were generally out of accord with the position.

Several points in the various compilations of case reports also are not in agreement with what we would expect were suffocation the entire story. For instance, Gafafer found the highest incidence not in the northern tier of states where bedding is at a maximum for the longest season but in the negroes of the south; New England, in fact, had the lowest rate. Abramson's finding that the peak is during the winter months is irrelevent in this respect since it might as well be correlated with the incidence of respiratory infections as the amount of bedding. Likewise the greater frequency in the economically less fortunate groups probably hooks up as well with crowding and poor living standards as with the possibility of mechanical compression since the same writer found evidence of overlaying in only a few instances. The mean age of 3 months for Abramson's cases would appear to be a little old if plain mechanical suffocation is to explain the picture since the 3- and 4- month baby is certainly robust enough to express vocally his views on other causes of discomfort and would, on the surface at least, be a less likely victim than a younger infant.

If we are led to doubt the importance of suffocation as an explanation of sudden death in sleeping infants, what other hypotheses can be substituted? The thymus has ceased to be popular although we would prefer it to suffocation; it implied no negligence nor did it infer that death was avoidable. In general, three groups of possibilities deserve consideration whenever the cause of death

*The implication of negligence when an infant dies from "suffocation" is so thoroughly ingrained that Templeman,[3] expressing the enlightened social outlook of the 1890's, suggested criminal prosecution of parents as one method of control. This despite the statement that, "Most of these cases occur in one-roomed houses, where frequently father, mother, and two, three, or even as many as five children sleep in one bed, which sometimes consists of a few jute sacks spread out on the floor."

in an infant is not obvious and regardless of whether the event occurred while asleep or awake. The first is infection by a variety of pathogens, the second is congenital anomalies, and the third might be called physiologic dysfunction.

Farber[4] was one of the first to emphasize overwhelming streptococcic septicemia in the etiology of unexpected death and recently we have seen two instances of meningococcemia terminate a matter of hours after the first symptom. The cause of death in one of these patients was not clear even at autopsy and was elicited only through a positive blood culture. A large part of our clinical evidence of infection is due to reactions by the body against the insulting agent, not to the presence of the organism per se, and it is of interest to speculate on the likelihood of rapid and relatively asymptomatic deterioration from septicemia in individuals virginal so far as previous conditioning of reaction is concerned. Topley[5] points out much the same problem in discussing local response to invasion. The various neurotropic viruses hold great theoretic potentiality as producers of sudden death because of their predilection for small cell collections essential to life. They notoriously produce little gross change which would be recognized during most post-mortem examinations. Goldbloom and Wiglesworth[6] have stressed the frequency with which they encountered a diffuse pneumonic process following unexplained death, but this might be another manifestation of an overwhelming septicemia rather than the actual cause of death; much the same relationship as exists between acute tracheobronchitis and the collapse which often accompanies it.

Congenital anomalies at a variety of strategic sites are capable of giving rise to sudden organic failure and death. The acyanotic type of cardiac defects may go along without recognition until the infant, through increasing body bulk and activity, provides a load sufficient to produce decompensation. Eight examples of this sequence have been presented by Levinson.[7] Rupture of congenital aneurysms, situated in such important areas as the base of the brain, can produce fatality as rapidly as can vascular accidents in older age groups, where the term suffocation is rarely used to explain the event. Mild degrees of laryngeal obstruction, sufficient to cause only periodic stridor during normal health, may be of extreme importance when edema from an otherwise mild respiratory infection is superimposed.

Physiologic dysfunction, while less clearly definable than the preceding categories, is one where speculation might prove most fruitful. The rapid growth, so characteristic of the early months, results in constant change of physiologic relationship and in gross instabilities; the exaggerated pyrexia in seemingly unimportant infections, the ease with which normal fluid and electrolyte interplay deteriorates during periods of minimal insult, the notoriously poor tolerance to surgery and anesthesia, and the rapidity with which vitamin deficiencies become manifest, are all reflections of the stage of disequilibrium. There must be a variety of these functional patterns capable of leading to sudden death. For example, hyperplasia of the isles of Langhans was the principal finding in a recently reported instance[6]—how often would hypoglycemia be detected clinically in the age group and how often is a careful microscopic study made routinely of the pancreas? Another writer,[8] in an attempt to explain anesthetic deaths, has postulated a development stage in infancy during which the respiratory center is markedly unstable to minimal changes in oxygen and carbon dioxide tension; possibly some such mechanism is of general importance.

These are by no means complete lists of the causes of sudden death in infancy but they illustrate some of the conditions to be considered before resort to suffocation. We, as pediatricians, should take the the lead in being a little more critical of the diagnosis and should reserve it for instances when actual deprivation of oxygen by mechanical means can be proved. Complete autopsy examination with full recourse to modern bacteriologic techniques should be demanded in every unexplained death of an infant. When this is done and still nothing is found, nor has incontrovertible evidence of suffocation been elicited, perhaps we should be pushed so far as to admit that we are ignorant of the cause of death, thereby saving the family the stigma of having allowed their baby to smother in the bedclothes. Status thymicolymphaticus or generalized incompatibility with life are more merciful and no more out of accord with the facts now available. We are heartily in accord with Dr. Abramson's belief that attention should be directed to the exceedingly large number of deaths which occur in supposedly healthy infants but we feel that further evidence is necessary before accounting the entire picture to mechanical suffocation.

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